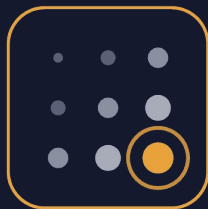


SMART CITY CROWD SAFETY PLATFORM



CROVIA

Turning the mobile network into a proactive crowd safety sensor — before the next crush, not after.

“Less Crowder More Safer”

A Fatal Blind Spot

Current crowd safety infrastructure is fundamentally reactive

1

Traditional CCTV cameras have blind spots and require constant monitoring. When a crowd shifts from a stadium into a narrow 6-metre street, a fatal crush can develop in under 4 minutes — faster than any human operator can respond.

2

Communication collapse: during surges, local cell towers become congested. First responders lose connectivity exactly when they need it most — delaying dispatch and coordination.

3

The bottleneck is not the crowd. It is the absence of a proactive, network-level safety layer.

Autonomous AI Crowd Disaster Prevention

1

The Insight

The mobile network already knows where every device is. We turn it into a crowd safety sensor — detecting surges before they become disasters.

2

Privacy-First

Opt-in consent via Smart City apps. All identifiers hashed. Aggregate signals used first — individual location retrieved only when a critical danger threshold is crossed.

3

Autonomous Action

When a critical Danger Score is reached, the AI simultaneously reroutes citizens via SMS, activates QoD network slices for first responders, and transmits GPS coordinates to emergency dispatch.

4

Micro-Rewards

Citizens acting as active probes or following rerouting instructions earn telecom perks — free 5G data or Smart City transit points — driving high opt-in rates.

Nokia NaC / GSMA Open Gateway APIs

Four APIs. One Safety Grid.

Dual-Trigger Architecture

Congestion Insights API tracks aggregate cell tower load — a privacy-safe primary trigger requiring no personal data. A spike means a crowd is forming. Geofencing API counts boundary crossings as a secondary trigger, eliminating blind spots when crowds migrate within one district. Location Retrieval API (consent-gated) pinpoints the crowd centroid. Quality on Demand (QoD) API reserves a guaranteed 5G network slice for first responder devices the moment a critical threshold is crossed.

1

Congestion Insights API

Aggregate cell tower load — privacy-safe primary trigger, no personal data.

2

Geofencing API

Boundary-crossing counts as a secondary trigger — eliminates blind spots.

3

Location Retrieval API

Consent-gated — pinpoints the crowd centroid.

4

Quality on Demand (QoD) API

Reserves a guaranteed 5G network slice for first responders.

Technical Architecture

1

Webhook & Consent Backend

Passively receives Geofencing and Congestion webhooks. Verifies user consent and hashes all PII before any data moves forward.

2

Spatial Clustering Engine

Runs DBSCAN on anonymized coordinates to calculate the GPS centroid of the crowd crush — offloaded from the AI core for speed.

3

LangGraph + Groq Agent

Llama 3 via Groq for ultra-low-latency reasoning. Queries Chroma RAG for infrastructure constraints, calculates Danger Score, triggers QoD sessions and SMS rerouting autonomously.

Ready Market, Clear Gap

The Market Is Ready

MEA smart-city investment projected to exceed \$30B by 2026. MENA IoT/software segment growing at 10.5% CAGR to \$3.95B by 2029. GSMA Open Gateway is already live across 292 networks — the infrastructure exists. The network-level safety layer does not.

The Gap We Fill

Saudi Arabia's 2026 Hajj season confirmed strong government appetite for AI crowd-safety spend — currently led by camera and drone systems, not network-level sensing. Egypt's New Administrative Capital is our pilot target: a greenfield smart city with no legacy safety infrastructure.

Market Sizing

Illustrative sizing across MENA smart-city and mega-event safety spend

Tier	Definition	Illustrative Scope
TAM	Public-safety & crowd-management software/services spend across MENA smart cities and mega-events	MENA smart-city IoT/software segment ≈ \$2.6–\$4.0B (2025–2029)
SAM	Cities, event authorities and pilgrimage authorities in GCC + Egypt actively procuring AI-driven, telecom-integrated public-safety platforms	GCC + Egypt flagship giga-projects, stadiums, festivals, Hajj/Umrah
SOM (Yr 1–3)	Anchor pilot-to-paid contracts the Platform can realistically close	3–5 flagship city/event contracts + 1–2 MNO revenue-share agreements

Competitive Landscape & Differentiation

Saudi Arabia's 2026 Hajj season already runs sophisticated AI crowd-safety systems — platforms developed with SDAIA use computer vision, thermal imaging and satellite/drone data to monitor density in real time. That confirms both budget and political appetite for this category, and shows the Platform's natural position: a complementary sensing and response layer, not a head-on camera replacement.

Dimension	CCTV / Computer-Vision Platforms	Dedicated IoT Sensor Networks	This Platform
Sensing method	Fixed cameras; visual blind spots	New hardware installs; high capex	Existing mobile network; no new hardware
Coverage	Limited to camera field of view	Limited to sensor placement	Anywhere with cellular coverage
Privacy model	Often continuous visual surveillance	Anonymous but hardware-bound	Opt-in consent + privacy-safe aggregate triggers
Response	Human-in-the-loop decision	Alerting only	Autonomous reroute + dispatch + emergency network slice
Deployment speed	Slow — physical installation	Slow — physical installation	Fast — software and API integration

B2G Software Licenses

1

Municipal Platform License

\$150K–\$500K per year. Annual fee scaled by coverage area and population.
No hardware capex — pure software delivery.

2

Mega-Event License

\$25K–\$150K per event. Scaled by peak attendance. Fastest to close — bridges revenue while government procurement runs.

3

Pilgrimage & National Security

\$500K–\$2M+ per season. Custom enterprise contract with Ministry of Interior or Hajj authority. Target gross margin 60–70% at scale.

Three-Phase Expansion



Team Roles & Bios

Ahmed El Mansy

Business Model & Monetization · Building Orchestrator & Connecting with Services

Worked on building the business model and monetization strategy and building orchestrator and connecting with services. Third year, Computer and Systems Engineering, Faculty of Engineering, Alexandria University.



Nour Mahmoud

Business Model & Monetization · Frontend & Design

Frontend and design and worked on building the business model and monetization. Third year, Computer and Systems Engineering, Faculty of Engineering, Alexandria University.



Ahmed Saied

Architecture & Model Selection · Integrating AI Layer with Webhooks

Integrating the AI layer with webhooks and helped decide the architecture services and models the platform runs on. Third year, Computer and Systems Engineering, Faculty of Engineering, Alexandria University.



Abdelrahman Ibrahim

Platform Vision & Architecture · AI Layer & Model Mind

Developed the platform's core vision and designed the architectural sequence mapping how the GSMA Open Gateway APIs integrate to enable autonomous safety triggers. Building AI layer and model mind. Third year, Computer and Systems Engineering, Faculty of Engineering, Alexandria University.



Mohamed Abdalla

Architecture · Testing & Debugging

Developed the platform's core vision and designed the architectural sequence mapping how the GSMA Open Gateway APIs integrate to enable autonomous safety triggers. Testing and debugging. Third year, Computer and Systems Engineering, Faculty of Engineering, Alexandria University.





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